In mathematics the art of proposing a question must be held of higher value than solving it.

Georg Cantor[1]

[1] **Georg Ferdinand Ludwig Philipp Cantor** (March 3, 1845, St. Petersburg, Russia – January 6, 1918, Halle, Germany), German mathematician who founded *set theory* and introduces the mathematically meaningful concept of *transfinite numbers*, indefinitely large but distinct from one another. He was the first person to take a systematic look at infinity. In fact, Cantor theorem implies the existence of an *infinity of infinities*.

National University of Computer and Emerging Sciences

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MT1004 Linear Algebra
BCS A – G

Sessional I Solution – Q1, Q2

Q1. a. Find two different row echelon forms of the matrix **(2)**

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

Solution: Let

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

(i).

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \overset{R_2 - 2R_1}{\rightarrow} \underbrace{\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}}_{Row\ Echelon\ Form} \overset{R_1 - 3R_2}{\rightarrow} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\substack{Row\ Reduced\ Echelon\ Form \\ (Not\ Required)}}.$$

(ii).

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \overset{R_2 \leftrightarrow R_1}{\rightarrow} \begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix} \overset{(1/2)R_1}{\rightarrow} \begin{bmatrix} 1 & 7/2 \\ 1 & 3 \end{bmatrix} \overset{R_2 - R_1}{\rightarrow} \begin{bmatrix} 1 & 7/2 \\ 0 & -1/2 \end{bmatrix} \overset{(-2)R_2}{\rightarrow} \begin{bmatrix} 1 & 7/2 \\ 0 & 1 \end{bmatrix} \overset{R_1 - (7/2)R_2}{\rightarrow} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\substack{RREF \\ (Not\ Required)}}.$$

The two row echelon forms are:

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 7/2 \\ 0 & 1 \end{bmatrix}.$$

Note that the row echelon forms can be different from the ones presented here.

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Q1. b. Let **(2)**

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

Write $A\boldsymbol{x}$ as linear combination of columns of A.

Solution:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-3) \begin{bmatrix} 3 \\ 7 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

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Q1. c. Consider the augmented matrix $([A|\boldsymbol{b}])$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 5 & 7 \\ 5 & 10 & 13 & 18 \end{bmatrix}$$

So that the row reduced echelon form is given by

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(i). Write the parametric form of the solution of the system $A\boldsymbol{x} = \boldsymbol{b}$. **(4)**

Solution: From the row reduced echelon form, the corresponding equations are:

$$x_1 + 2x_2 = 0, \qquad x_3 = 1, \qquad 0 = 0\,.$$

Clearly $x_1 + 2x_2 = 0$ gives $x_1 = -2x_2$. The parametric form is given by

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\boldsymbol{x}} = \begin{bmatrix} -x_2 \\ x_2 \\ 1 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

or

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{\boldsymbol{v}} \underbrace{x_2}_{t} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\boldsymbol{p}}.$$

The compact expression of the solution is

$$\boldsymbol{x} = t\boldsymbol{v} + \boldsymbol{p}\,.$$

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(ii). Write the solution of corresponding homogenous system $Ax = 0$. **(2)**

Solution: It is evident from parametric form that the solution of corresponding homogenous system $Ax = 0$ is $\boldsymbol{x} = t\boldsymbol{v}$. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} x_2\,.$$

(iii). What is the geometric relation between the two solutions. **(2)**

Solution: Geometrically speaking, we say that $\boldsymbol{v}$ is translated by $\boldsymbol{p}$ to $\boldsymbol{v} + \boldsymbol{p}$. That is, the solution of $A\boldsymbol{x} = \boldsymbol{0}$ is translated to the solution of $A\boldsymbol{x} = \boldsymbol{b}$ through vector $\boldsymbol{p}$.

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Q1. d. Explain why any basis of $\mathbb{R}^3$ can't have more or less than three vectors. **(1)**

Solution: In case the number of vectors, say m, is more than three ($m > 3$), the set of vectors become linearly dependent, a violation. When ($m < 3$), the *span* of vectors is not the complete space $\mathbb{R}^3$, a violation.

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Q2. a. Explain why the solution of $A\boldsymbol{x} = \boldsymbol{b}$ is not a subspace. **(2)**

Solution: The solution of $A\boldsymbol{x} = \boldsymbol{b}$ do not contain the zero vector $\mathbf{0}$**,** one of the primary requirements for a subset to be a subspace.

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Q2. b. Let

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & \alpha \end{bmatrix}.$$

(i). Find the value(s) of α for which the columns of the given matrix are linearly independent. **(2)**

(ii). Find the value(s) of α, the span of columns of A is a line. **(2)**

Solution:

(i). Method I: By Checking For Linear Independence Systematically

To check linear independence for columns of A, one requires to solve $A\boldsymbol{x} = \mathbf{0}$. The corresponding augmented matrix is $[A|\mathbf{0}]$.

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 2 & 4 & \alpha & 0 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & 2 & 4 & 0 \\ 0 & 0 & \alpha - 8 & 0 \end{bmatrix}.$$

When $\alpha - 8 = 0$, implies $\alpha = 8$, the columns of A are linearly dependent.

For $\alpha \neq 8$, we have

$$x_1 + 2x_2 + 4x_3 = 0,$$

and

$$x_3 = 0.$$

When $x_3 = 0$, $x_1 + 2x_2 + 4x_3 = 0$ can be written as

$$x_1 = -2x_2 .$$

Then

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} x_2 .$$

Since the solutions are non-trivial ($\boldsymbol{x} \neq \mathbf{0}$) the columns will remain linearly dependent even for $\alpha \neq 8$.

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Method II: By Inspection

By inspection, the first and second columns of given matrix are linearly dependent, implies that for all values of α the columns of matrix will be linearly dependent.

(ii). For $\alpha = 8$, A can be written as:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}.$$

Then

$$Col\ A = span\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix}. \begin{bmatrix} 4 \\ 8 \end{bmatrix}\right\} = span\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right\} \cong \mathbb{R}, a\ line .$$

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Q2. c. Find the equation of plane spanned by the following set of vectors. **(4)**

$$\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\7\\9\end{bmatrix}\right\}.$$

Solution: Method I: Using Approach of Multivariable Calculus.

Let

$$\boldsymbol{v_1} = \begin{bmatrix}1\\2\\3\end{bmatrix}, \qquad \boldsymbol{v_2} = \begin{bmatrix}2\\7\\9\end{bmatrix}.$$

The equation of plane given a point $P_0(x_0, y_0, z_0)$ and a normal vector $\boldsymbol{n} = (a, b, c)$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0\,.$$

Since we have span of two vectors which contains the $\mathbf{0}$ vector, one can choose P_0 to be $(0,0,0)$. The equation of plane can then be written as

$$ax - by + cz = 0\,.$$

That is, one needs the normal vector $\boldsymbol{n}$ in this case. As a matter of fact, $\boldsymbol{n}$ is a normal vector to the plane implies

$$\boldsymbol{n} \cdot \boldsymbol{v_1} = 0,$$

and

$$\boldsymbol{n} \cdot \boldsymbol{v_2} = 0\,.$$

Implies

$$a + 2b + 3c = 0, \qquad 2a + 7b + 9c = 0\,.$$

The matrix form is

$$\begin{bmatrix}1 & 2 & 3\\2 & 7 & 9\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}.$$

The augmented matrix $[A|b]$ is

$$\begin{bmatrix}1 & 2 & 3 & 0\\2 & 7 & 9 & 0\end{bmatrix}.$$

Now

$$\begin{bmatrix}1 & 2 & 3 & 0\\2 & 7 & 9 & 0\end{bmatrix} \overset{R_2 - 2R_1}{\rightarrow} \begin{bmatrix}1 & 2 & 3 & 0\\0 & 3 & 3 & 0\end{bmatrix} \overset{(1/3)R_2}{\rightarrow} \begin{bmatrix}1 & 2 & 3 & 0\\0 & 1 & 1 & 0\end{bmatrix} \overset{R_1 - 2R_2}{\rightarrow} \begin{bmatrix}1 & 0 & 1 & 0\\0 & 1 & 1 & 0\end{bmatrix}.$$

This gives

$$a + c = 0,$$

and

$$b + c = 0, \qquad c = -b\,.$$

Plugging $c = -b$ in $a + c = 0$ gives $a - b = 0$. Implies $a = b$.

The parametric form of solution then can be written as:

$$\begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}b\\b\\-b\end{bmatrix} = \begin{bmatrix}1\\1\\-1\end{bmatrix} b = span\left\{\begin{bmatrix}1\\1\\-1\end{bmatrix}\right\}.$$

By choosing for different values of b we have a normal vector to the plane. For instance, if $b = 1$, then

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix},$$

which gives $ax + by + cz = 0$ as

$$x + y - z = 0\,.$$

Say if we choose any other value of b, e.g. $b = 2$. Then

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix},$$

which gives $ax + by + cz = 0$ as

$$2x + 2y - 2z = 0,$$

or

$$x + y - z = 0\,.$$

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Method II: Using Definition of Span.

Let $\boldsymbol{u} = (x, y, z)$ be a point in $span\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, then there exist $s, t \in \mathbb{R}$ such that

$$\boldsymbol{u} = s\boldsymbol{v_1} + t\boldsymbol{v_2}\,.$$

Implies

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = s\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t\begin{bmatrix} 2 \\ 7 \\ 9 \end{bmatrix},$$

Or

$$\begin{bmatrix} 1 & 2 \\ 2 & 7 \\ 3 & 9 \end{bmatrix}\begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

The augmented matrix is $[A|b]$ is

$$\begin{bmatrix} 1 & 2 & x \\ 2 & 7 & y \\ 3 & 9 & z \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & 2 & x \\ 0 & 3 & y - 2x \\ 0 & 3 & z - 3x \end{bmatrix} \begin{matrix} R_3 - R_2 \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & 2 & x \\ 0 & 3 & y - 2x \\ 0 & 0 & z - 3x - y + 2x \end{bmatrix} \begin{matrix} \left(\frac{1}{3}\right)R_2 \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & 2 & x \\ 0 & 1 & \frac{1}{3}(y - 2x) \\ 0 & 0 & -x - y + z \end{bmatrix} \begin{matrix} \left(\frac{1}{3}\right)R_2 \\ \rightarrow \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & x - \frac{2}{3}(y - 2x) \\ 0 & 1 & \frac{1}{3}(y - 2x) \\ 0 & 0 & -x - y + z \end{bmatrix}.$$

The above system has a solution if and only if

$$-x - y + z = 0,$$

or

$$x + y - z = 0,$$

which is required equation of the plane.

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